$\mu = 63.67/75, \quad \sigma = 10.59$

ECE 329 Fields and Waves I Summer 17

Kudeki

Exam 3

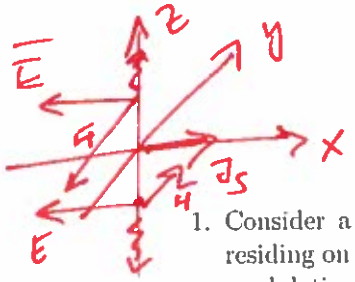
Fri, July 28, 2017 — 11 AM - 11:50

Name:	Solution
Section:	

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the above boxes.

This is a closed book exam and calculators are allowed. You are allowed three formula sheets of 8.5 by 11 inch dimensions — both sides of the sheets may be used. Please show all your work and make sure to include your reasoning for each answer. All answers should include units wherever appropriate.

Problem 1 (25 points)	
Problem 2 (25 points)	
Problem 3 (25 points)	
TOTAL (75 points)	



1. Consider a time-varying surface current density $\mathbf{J}_s(t) = u(t)\sin(2\pi ft)\hat{x}$ A/m, with $f = 10^6$ Hz, residing on the $z = 0$ surface embedded in free space. In this expression $u(t)$ is the unit-step function modulating the sine signal with a frequency $\omega = 2\pi f$ rad/s.

a) (4 pts) What is the vector wave field $\mathbf{E}(z,t)$ caused by $\mathbf{J}_s(t)$ in the region $z > 0$?

$$\bar{E}(z,t) = -\hat{x}\,\frac{\eta_0}{2}\,u\left(t-\frac{z}{c}\right)\sin\left(2\pi f\left(t-\frac{z}{c}\right)\right),\quad z>0$$

b) (4 pts) What is the accompanying wavefield $\mathbf{H}(z,t)$ in the same region?

$$\bar{H}(z,t) = -\hat{y}\,\frac{1}{2}\,u\left(t-\frac{z}{c}\right)\sin\left(2\pi f\left(t-\frac{z}{c}\right)\right),\quad z>0$$

c) (3 pts) What is the period $T = \frac{2\pi}{\omega}$ of the modulated $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$ waveforms?

$$T = \frac{1}{f} = 10^{-6}\,s = 1\,\mu s$$

d) (3 pts) What is the wavelength λ of the modulated $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$ waveforms?

$$\lambda = \frac{c}{f} = \frac{3\times 10^8\ m/s}{10^6\ 1/s} = 300\ m$$

e) (5 pts) Plot $E_x(z,t)$ vs t at $z = 600$ m over the time interval $0 < t < 4\ \mu s$. Label both axes carefully. **Hint:** E_x is initially zero until the wavefront arrives at 600 m.

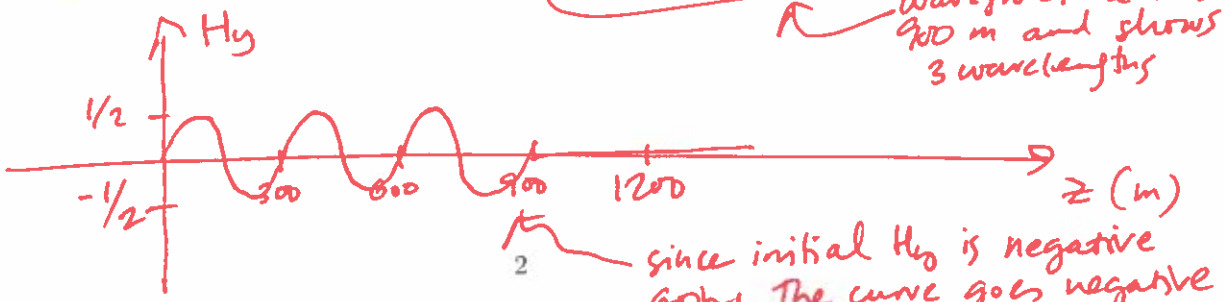
at 2 μs



a delayed minus sine curve with 1 μs period !!

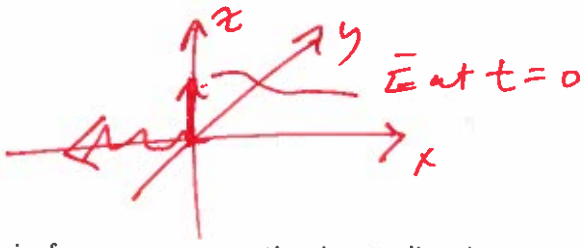
f) (6 pts) Plot $H_y(z,t)$ vs z at $t = 3\,\mu s$ over the region $0 < z < 1200$ m. Label both axes carefully. **Hint:** H_y is zero at distances where the wavefront has not yet reached.

wavefront reaches 900 m and shows 3 wavelengths



since initial H_y is negative going the curve goes negative at the wavefront at 900 m.

$\lambda = \frac{c}{f} = \frac{3\times10^8\ m/s}{100\times10^6\ 1/s} = 3\ m$



2. This problem has two independent parts:

a) Consider a *left*-circular-polarized TEM plane wave in free-space propagating in $-x$ direction. The instantaneous electric field vector of the wave is $20\hat{z}$ V/m at time $t = 0$ at the location $(x, y, z) = (0, 0, 0)$. The wave frequency is $f = 100$ MHz. If the wave electric field phasor is denoted as

$$\tilde{\mathbf{E}} = (\mathcal{A}\hat{x} + \mathcal{B}\hat{y} + \mathcal{C}\hat{z})e^{j(ax+by+cz)}\ \frac{\text{V}}{\text{m}}$$

determine:

i. (4 pts) The numerical values of constants a, b, and c. **Hint:** what is β?

$b = c = 0, \quad a = \beta = \frac{2\pi}{\lambda} = \frac{2\pi}{3}\ \frac{rad}{m}$ ///

ii. (4 pts) The numerical values of constants $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$.

E_y leads E_z so that the wave LHP

$\mathcal{C} = 20\ \frac{V}{m} \Rightarrow \mathcal{B} = +j20\ \frac{V}{m}$ & $\mathcal{A} = 0$.

iii. (4 pts) The time-averaged Poynting vector $\langle \mathbf{E} \times \mathbf{H} \rangle$. **Hint:** twice the linear polarized value

$\langle \bar{E}\times\bar{H} \rangle = -\hat{x}\,2\times\frac{|\mathcal{C}|^2}{2\eta_0} = -\hat{x}\,\frac{20^2}{120\pi} = -\hat{x}\,\frac{10}{3\pi}\ W/m^2$

b) Consider TEM wave propagation in some homogeneous medium — the wave frequency is $f = 100$ MHz, wavelength $\lambda = 2$ m, penetration depth $\delta = 100$ m, and $\mu = \mu_o$.

i. (4 pts) Does the medium behave as an *imperfect dielectric* or else as a *good conductor* at the given wave frequency? Explain why.

$\lambda << \delta \Rightarrow \beta = \frac{2\pi}{\lambda} >> \alpha = \frac{1}{\delta} \Leftrightarrow$ Imperfect dielectric

ii. (2 pts) What is the numerical value of propagation velocity v?

$\lambda f = v = (2\ m)(100\times10^6\ \tfrac{1}{s}) = 2\times10^8\ \frac{m}{s}$ //

iii. (2 pts) What is the numerical value of relative permittivity ϵ_r?

$v = \frac{1}{\sqrt{\mu\epsilon}} \rightarrow \sqrt{\epsilon_r} = \frac{c}{v} \rightarrow \epsilon_r = \frac{c^2}{v^2} = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$

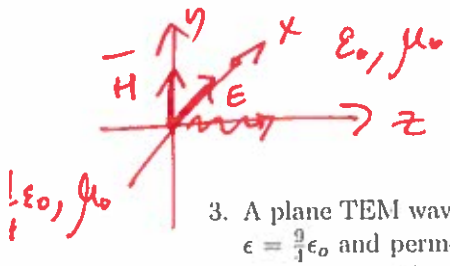
iv. (2 pts) What is the numerical value of the magnitude of intrinsic impedance η?

$\eta = \sqrt{\frac{\mu_0}{\epsilon_r \epsilon_0}} = \frac{\eta_0}{3/2} = \frac{2}{3}\,120\pi = 80\pi\ \Omega$

v. (3 pts) What is the numerical value of conductivity σ.

$\alpha = \beta\frac{\sigma}{2\omega\epsilon} \Rightarrow \sigma = 2\omega\epsilon\frac{\alpha}{\beta} = 2\,2\pi f\,\frac{9}{4}\,\frac{1}{36\pi\,10^9}\,\frac{1}{\delta}\,\frac{\lambda}{2\pi}$

$= 2\times10^8\,\frac{9}{4}\,\frac{1}{36\pi\,10^9}\,\frac{1}{100}\,2 = \frac{10^{-3}}{4\pi}\ S/m$



3. A plane TEM wave is propagating $+z$ direction in a homogeneous perfect dielectric with permittivity $\epsilon = \frac{9}{1}\epsilon_o$ and permeability $\mu = \mu_o$ and is incident on the $z = 0$ plane which is the boundary between the perfect dielectric and vacuum. The "transmitted" electric field vector function found in vacuum in the $z > 0$ region is given as

$$\mathbf{E}_t(z,t) = 6\cos(\omega t - \beta z)\hat{x}\,\frac{\text{V}}{\text{m}}$$

This is in free space, $z>0$

where $f = \frac{\omega}{2\pi} = 1$ MHz. Determine (expressing all answers in simplified numerical rather than symbolic forms except for the π factors):

a) (3 pts) The intrinsic impedance η for both regions $z < 0$ and $z > 0$.

$z>0$ $\quad \eta = \eta_0 = 120\pi\ \Omega$

$z<0$ $\quad \eta = \sqrt{\frac{\mu_0}{\frac{9}{1}\varepsilon_0}} = \frac{2}{3}\eta_0 = 80\pi\ \Omega$ ///

b) (3 pts) The wavenumber β for both regions $z < 0$ and $z > 0$.

$\beta = \omega\sqrt{\mu\varepsilon} = \sqrt{\varepsilon_r}\,\frac{\omega}{c} \Rightarrow z<0,\ \beta = \frac{3}{2}\,\frac{2\pi\,10^6}{3\times10^8} = \frac{\pi}{100}$ rad/m ///

$z<0,\ \beta = 1\cdot\frac{2\pi\,10^6}{3\times10^8} = \frac{\pi}{150}$ rad/m ///

c) (3 pts) The reflection coefficient Γ for wave propagating in $+z$ direction.

$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{\eta_0 - \frac{2}{3}\eta_0}{\eta_0 + \frac{2}{3}\eta_0} = \frac{3-2}{3+2} = \frac{1}{5} = 0.2,\quad \tau = 1+\Gamma = 1.2$

d) (6 pts) The electric field amplitudes of the incident and reflected waves in the perfect dielectric.

$E_r + E_i = E_t \Rightarrow \underbrace{(1+\Gamma)}_{1.2} E_i = \underbrace{E_t}_{6} \Rightarrow E_i = \frac{6}{1.2} = 5$ V/m

$E_r = 5\times\frac{1}{5} = 1$ V/m

e) (4 pts) Phasor $\tilde{\mathbf{E}}_r(z)$ of the reflected wave field in region $z < 0$.

$\tilde{\bar{E}}_r(z) = \hat{x}\,1\,e^{+j\beta_1 z} = \hat{x}\,e^{j\frac{\pi}{100}z}$ V/m //.

f) (3 pts) Phasor $\tilde{\mathbf{H}}_t(z)$ of the transmitted wave field in region $z > 0$.

$\tilde{\bar{H}}_t(z) = \hat{y}\,\frac{6e^{-j\beta_2 z}}{\eta_0} = \hat{y}\,\frac{6}{120\pi}\,e^{-j\frac{\pi}{150}z} = \hat{y}\,\frac{1}{20\pi}\,e^{-j\frac{\pi}{150}z}$ A/m //.

g) (3 pts) The time-averaged Poynting vector $\langle \mathbf{E}_t \times \mathbf{H}_t \rangle$ of the transmitted wave.

$\langle \bar{E}\times\bar{H} \rangle = \hat{z}\,\frac{|\tilde{\bar{E}}_t|^2}{2\eta_0} = \hat{z}\,\frac{6\times6}{2\times120\pi} = \hat{z}\,\frac{3}{20\pi}$ W/m^2 //.